Anthropic

(Claude Certified Developer Foundations)

Claude Certified Developer Foundations (CCDV-F)

Total: **101 Questions**

Link: https://certyiq.com/papers/anthropic/ccdv-f

Question: 1 CertyIQ

A developer creates an AI coding assistant for a known workflow with repeatable steps. Which design approach is appropriate?

A. Use a fixed sequence of prompts
B. Allow unlimited autonomous exploration
C. Remove all workflow instructions
D. Use random decision paths

Answer: A

Explanation:

A. Use a fixed sequence of prompts

Correct

For a **known, repeatable workflow**, a fixed sequence of prompts provides predictable and consistent execution. It guides the AI assistant through each required step in the correct order.

B. Allow unlimited autonomous exploration

Incorrect

Unlimited exploration can make the assistant unpredictable and is unnecessary when the workflow is already well-defined.

C. Remove all workflow instructions

Incorrect

Without workflow instructions, the assistant may produce inconsistent results and fail to follow the required process.

D. Use random decision paths

Incorrect

Random decision paths are inappropriate for repeatable workflows because they reduce consistency and predictability.

Final Answer: A

A = Use a fixed sequence of prompts — this is the appropriate design for a known workflow with repeatable steps.

Question: 2 CertyIQ

A team wants stable automated pull request review results from Claude Code. Which method best supports automation?

A. Split responses using headings
B. Use stricter markdown parsing rules
C. Post the entire response for manual review
D. Use JSON output with a defined schema

Answer: D

Explanation:

A. Split responses using headings

Incorrect

Headings can improve readability, but they do not guarantee a consistent machine-readable structure for automated processing.

B. Use stricter markdown parsing rules

Incorrect

Markdown parsing can still be fragile because the response structure may vary. It is not ideal for reliable automation.

C. Post the entire response for manual review

Incorrect

Manual review does not provide stable, automated processing of pull request review results.

D. Use JSON output with a defined schema

Correct

A **defined JSON schema** provides a predictable and structured output format. Automation tools can reliably parse fields such as findings, severity, file, line, and recommendations, making it well suited for consistent PR review workflows.

Final Answer: D

D = Use JSON output with a defined schema — structured, schema-based output provides predictable results that can be reliably consumed by automation.

Question: 3 CertyIQ

A team needs to update every file ending with `.fixture.json` across a monorepo. Searching for the word “fixture” creates noisy results and misses sparse files.
Which tool-use strategy should be adopted?

A. Use Bash commands and paste the entire directory listing.
B. Use Glob patterns like `**/*.fixture.json` and then Read matching files.
C. Read all possible files before searching.
D. Use Grep for "fixture" and infer matching files.

Answer: B

Explanation:

A. Use Bash commands and paste the entire directory listing.

Incorrect

A full directory listing can be unnecessarily large and does not precisely identify the files that match the required filename pattern.

B. Use Glob patterns like **/*.fixture.json and then Read matching files.

Correct

Glob is designed for finding files based on filename patterns. Using **/*.fixture.json efficiently identifies all matching files across the monorepo, after which **Read** can inspect their contents before making updates.

C. Read all possible files before searching.

Incorrect

Reading all files is inefficient and consumes unnecessary context. The team should first identify the relevant files using a precise pattern.

D. Use Grep for "fixture" and infer matching files.

Incorrect

Grep searches file contents, not filenames. It can produce noisy results and miss files that do not contain the word fixture.

Final Answer: B

B = Use Glob patterns like **/*.fixture.json and then Read matching files — Glob precisely finds files by name/pattern, making it the appropriate tool for this task.

Question: 4 CertyIQ

Scenario: Developer Productivity with Claude
Your CI job invokes Claude Code successfully and posts a single pull request summary comment. The team now wants automated inline comments for each finding, but the integration frequently breaks because Claude sometimes adds prose, changes field names, or formats findings as markdown.
What change best supports reliable automation?

A. Parse Claude's markdown bullets with regular expressions.

B. Strengthen the prompt to demand valid JSON only.

C. Run Claude Code with `--output-format json` and `--json-schema`, then map validated findings to inline PR comments.

D. Store previous comments and ask Claude to rewrite them into records.

Answer: C

Explanation:

A. Parse Claude's markdown bullets with regular expressions.

Incorrect

Markdown parsing with regular expressions is fragile. Changes in formatting or wording can easily break the automation.

B. Strengthen the prompt to demand valid JSON only.

Incorrect

Prompt instructions alone do not guarantee reliable structured output. Claude may still add prose, change fields, or produce invalid JSON.

C. Run Claude Code with --output-format json and --json-schema, then map validated findings to inline PR comments.

Correct

Using **structured JSON output with a defined schema** provides predictable field names and a machine-readable format. The CI workflow can validate the findings and reliably map each result to the appropriate inline pull request comment.

D. Store previous comments and ask Claude to rewrite them into records.

Incorrect

Reprocessing previous comments introduces unnecessary complexity and still does not guarantee a stable, validated structure for automation.

Final Answer: C

C = Use --output-format json with --json-schema and map the validated findings to inline PR comments — this provides reliable, structured output for CI automation.

Question: 5 CertyIQ

Scenario: Multi-Agent Research System
The document analysis subagent receives coordinator-selected PDFs and reports from an approved catalog, but it has a generic URL retrieval tool.
It sometimes follows links inside documents to:
l blogs
l login pages
l duplicate HTML summaries and then cites those pages instead of the approved sources.
What change best addresses this?

A. Keep fetch_url available, but add prompt instructions warning the subagent never to open links found inside documents.

B. Replace fetch_url with a load_document tool that accepts catalog document IDs or approved URLs and validates before fetching.

C. Give the document analysis subagent web search tools too, so it can independently confirm whether linked pages are relevant.

D. Allow fetch_url for any link, then have synthesis discard citations whose domains are not in the approved catalog.

Answer: B

Explanation:

A. Keep fetch_url available, but add prompt instructions warning the subagent never to open links found inside documents.

Incorrect

Prompt instructions can help guide behavior, but they do not technically prevent the subagent from accessing unapproved URLs. A tool-level restriction is more reliable.

B. Replace fetch_url with a load_document tool that accepts catalog document IDs or approved URLs and validates before fetching.

Correct

A constrained **load_document tool** enforces the approved-source boundary at the tool level. By accepting only catalog document IDs or validated approved URLs, the subagent cannot freely follow arbitrary links found inside documents.

C. Give the document analysis subagent web search tools too, so it can independently confirm whether linked pages are relevant.

Incorrect

This expands the subagent's access to external sources and increases the possibility of citing unapproved content.

D. Allow fetch_url for any link, then have synthesis discard citations whose domains are not in the approved catalog.

Incorrect

This allows the document analysis agent to retrieve and potentially rely on unapproved information before synthesis filters the citations. Prevention at the retrieval layer is safer and more reliable.

Final Answer: B

B = Replace fetch_url with a validated load_document tool — enforce the approved-source boundary at the tool level rather than relying only on prompts or downstream filtering.

Question: 6

CertyIQ

A coordinator launches subagents to analyze API handlers, database writes, and scheduled jobs. Final architecture reports often attach claims to incorrect files or omit line ranges.
What improves handoff reliability?

A. Concatenate complete transcripts
B. Use regex parsing on summaries.
C. Ask the synthesizer to reconstruct locations.
D. Require structured handoff records with paths, symbols, line ranges, and evidence snippets.

Answer: D

Explanation:

A. Concatenate complete transcripts.

Incorrect

Complete transcripts can contain large amounts of irrelevant information and do not guarantee that file locations and evidence are represented consistently.

B. Use regex parsing on summaries.

Incorrect

Regex parsing of free-form summaries is fragile. Changes in wording or formatting can cause locations and evidence to be extracted incorrectly.

C. Ask the synthesizer to reconstruct locations.

Incorrect

The synthesizer should not have to guess or reconstruct file locations. This can introduce incorrect paths, symbols, or line ranges.

D. Require structured handoff records with paths, symbols, line ranges, and evidence snippets.

Correct

Structured handoff records preserve **precise, machine-readable evidence** from each subagent. Including paths, symbols, line ranges, and evidence snippets allows the synthesizer to accurately connect architectural claims to the correct source locations.

Final Answer: D

D = Require structured handoff records with paths, symbols, line ranges, and evidence snippets — structured evidence makes multi-agent handoffs precise, traceable, and reliable.

Question: 7

CertyIQ

Scenario: Structured Data Extraction
Your QA pipeline flags mistakes in supplier contracts.
Analysts accept many findings about missing renewal clauses but reject most findings involving informal pricing language.
Current logs store only:
l Document ID
l Field name
l Confidence
l Finding text
What change should you make first?

A. Increase QA confidence threshold globally.

B. Add a structured detected_pattern field to each QA finding, then analyze acceptance rates by pattern and document type.

C. Store only analyst decisions.

D. Tell Claude to report only findings analysts accept.

Answer: B

Explanation:

A. Increase QA confidence threshold globally.

Incorrect

A global confidence threshold would affect all findings and could incorrectly remove useful renewal-clause findings. The problem appears to depend on the **type of detected pattern**, not simply confidence.

B. Add a structured detected_pattern field to each QA finding, then analyze acceptance rates by pattern and document type.

Correct

Adding a structured **detected_pattern** field makes findings easier to categorize and analyze. The team can determine which patterns — such as missing renewal clauses or informal pricing language — have higher or lower acceptance rates and then improve the system based on evidence.

C. Store only analyst decisions.

Incorrect

Storing only acceptance/rejection decisions loses important information about what pattern generated the finding, making it difficult to identify why certain findings are rejected.

D. Tell Claude to report only findings analysts accept.

Incorrect

This would hide potentially useful findings rather than improving the detection process. It also prevents the team from analyzing rejected findings and understanding systematic errors.

Final Answer: B

B = Add a structured detected_pattern field and analyze acceptance rates by pattern and document type — this provides the data needed to identify systematic QA errors and improve extraction quality.

Question: 8 CertyIQ

A multi-agent system uses a coordinator and specialized researchers. How should required MCP server configuration be shared across a team repository?

A. Ask every user to recreate servers manually
B. Store all servers in personal user configuration
C. Combine all integrations into one MCP server
D. Place required shared servers in project-scoped .mcp.json

Answer: D

Explanation:

A. Ask every user to recreate servers manually

Incorrect

Manual recreation is inconsistent and creates unnecessary setup work. Team members may configure servers differently, leading to inconsistent agent behavior.

B. Store all servers in personal user configuration

Incorrect

Personal configuration is tied to an individual user and is not appropriate for sharing required project-level configuration with the entire team.

C. Combine all integrations into one MCP server

Incorrect

Combining unrelated integrations into a single server reduces modularity and is unnecessary. MCP servers

can remain separate while being centrally configured for the project.

D. Place required shared servers in project-scoped .mcp.json

Correct

A project-scoped **.mcp.json** allows required MCP server configuration to be stored with the repository and shared consistently across the team. This ensures the coordinator and specialized agents can use the required integrations with a standardized setup.

Final Answer: D

D = Place required shared servers in project-scoped .mcp.json — this provides consistent, version-controlled MCP configuration that can be shared across the team repository.

Question: 9 CertyIQ

A customer support agent alternates unpredictably between `get_customer` and `lookup_order` when handling mixed requests. The MCP tool catalog only contains short autogenerated descriptions.
What change should you make first to improve tool selection reliability?

A. Update the system prompt to tell Claude to think carefully before selecting tools.

B. Add a keyword router that maps phrases like “order” or “refund” to one tool.

C. Consolidate get_customer and lookup_order into one broad lookup tool.

D. Rewrite each MCP tool description to specify purpose, accepted identifiers, returned fields, edge cases, and boundaries.

Answer: D

Explanation:

A. Update the system prompt to tell Claude to think carefully before selecting tools.

Incorrect

Simply telling Claude to think carefully does not provide the detailed information needed to distinguish between similar tools. Better tool metadata is more effective.

B. Add a keyword router that maps phrases like “order” or “refund” to one tool.

Incorrect

Keyword routing is brittle because real customer requests can contain multiple concepts or ambiguous language. It can also bypass the model's understanding of the tools' actual capabilities.

C. Consolidate get_customer and lookup_order into one broad lookup tool.

Incorrect

Combining the tools reduces specialization and can make the tool interface less clear. It does not address the underlying problem of insufficient tool descriptions.

D. Rewrite each MCP tool description to specify purpose, accepted identifiers, returned fields, edge cases, and boundaries.

Correct

Detailed tool descriptions give Claude the information needed to **differentiate similar tools and select the appropriate one**. Clearly defining each tool's purpose, inputs, outputs, edge cases, and boundaries improves reliable tool selection.

Final Answer: D

D = Rewrite each MCP tool description with clear purpose, identifiers, returned fields, edge cases, and boundaries — high-quality tool descriptions are critical for reliable MCP tool selection.

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Question: 11 CertyIQ

A debugging agent investigates unfamiliar systems where the correct solution is unknown at the beginning. Which strategy should it use?

A. Follow a completely fixed workflow
B. Use adaptive planning based on discoveries
C. Ignore intermediate findings
D. Always restart from the beginning

Answer: B

Explanation:

A. Follow a completely fixed workflow

Incorrect

A completely fixed workflow works best for predictable, repeatable tasks. In unfamiliar systems, new discoveries may require the agent to change its approach.

B. Use adaptive planning based on discoveries

Correct

An unfamiliar debugging problem requires the agent to **inspect findings, update its understanding, and adjust the plan dynamically**. Adaptive planning allows the agent to respond to new evidence and pursue the most promising path toward a solution.

C. Ignore intermediate findings

Incorrect

Intermediate findings provide important evidence about the system and should be used to guide subsequent debugging steps.

D. Always restart from the beginning

Incorrect

Restarting from scratch wastes previous discoveries and makes debugging less efficient.

Final Answer: B

B = Use adaptive planning based on discoveries — when the correct solution is initially unknown, the agent should continuously update its plan based on new evidence.

Question: 12 CertyIQ

An engineer asks Claude to add a caching layer around a legacy authorization module. The system has multiple tenants, audit logging, and inconsistent invalidation behavior.

What should happen before modifying files?

A. Implement a simple cache wrapper and iterate later.
B. Ask targeted questions about invariants, invalidation, audit requirements, and failure modes.
C. Provide performance goals and let Claude infer rules.
D. Copy caching patterns from another module.

Answer: B

Explanation:

A. Implement a simple cache wrapper and iterate later.

Incorrect

Implementing immediately could introduce serious issues in authorization, tenant isolation, audit logging, or cache invalidation. These requirements should be clarified first.

B. Ask targeted questions about invariants, invalidation, audit requirements, and failure modes.

Correct

Before modifying a complex legacy system, Claude should clarify the **security and operational constraints**. Targeted questions help identify authorization invariants, tenant boundaries, invalidation rules, audit requirements, and failure scenarios before implementation begins.

C. Provide performance goals and let Claude infer rules.

Incorrect

Performance goals alone do not define critical authorization and security requirements. Letting Claude infer these rules could result in unsafe assumptions.

D. Copy caching patterns from another module.

Incorrect

Another module may have different authorization, tenant, or invalidation requirements. Reusing its pattern without validating the assumptions could introduce defects.

Final Answer: B

B = Ask targeted questions about invariants, invalidation, audit requirements, and failure modes — clarify critical constraints before making changes to a complex legacy authorization system.

Question: 13 CertyIQ

A developer creates a multi-agent workflow where one agent researches documentation and another writes implementation plans. How should information flow between agents be handled?

A. Assume all agents share the same conversation context
B. Pass required findings through the coordinator
C. Allow each agent to access every previous message
D. Remove the coordinator role

Answer: B

Explanation:

A. Assume all agents share the same conversation context

Incorrect

Agents should not be assumed to automatically share the same conversation context. Each agent may have its own context and scope.

B. Pass required findings through the coordinator

Correct

The **coordinator** should collect relevant findings from the research agent and pass the necessary information to the planning agent. This creates a controlled and reliable information flow between specialized agents.

C. Allow each agent to access every previous message

Incorrect

Giving every agent access to all previous messages increases unnecessary context and can reduce control over what information each agent receives.

D. Remove the coordinator role

Incorrect

The coordinator provides orchestration and controls how information and tasks move between specialized agents. Removing it makes the workflow harder to manage reliably.

Final Answer: B

B = Pass required findings through the coordinator — the coordinator manages controlled information handoffs between specialized agents.

Question: 14 CertyIQ

A structured extraction system divides documents into sections and sends them to extraction subagents. One subagent temporarily loses access to its source system. The current implementation either returns an empty successful extraction or aborts the entire document.
What should you change?

A. Return an empty extraction object for the failed section.

B. Terminate the entire document workflow whenever any subagent fails.

C. Have subagents report unresolved failures with attempted section, failure type, partial fields, and recovery suggestions.

D. Retry the failed section until it succeeds before returning anything.

Answer: C

Explanation:

A. Return an empty extraction object for the failed section.

Incorrect

An empty extraction can incorrectly suggest that the section contained no relevant information. It hides the actual failure and makes downstream results less trustworthy.

B. Terminate the entire document workflow whenever any subagent fails.

Incorrect

A single subagent failure should not necessarily cause the entire document workflow to fail. Other sections may still be successfully processed.

C. Have subagents report unresolved failures with attempted section, failure type, partial fields, and recovery suggestions.

Correct

Subagents should return **structured failure information** rather than silently producing empty results. Reporting the attempted section, failure type, partial results, and recovery suggestions allows the coordinator to distinguish genuine empty results from unresolved failures and decide how to recover.

D. Retry the failed section until it succeeds before returning anything.

Incorrect

Indefinite retries can block the entire workflow and may not resolve a persistent source-system failure. Failures should be reported explicitly so the workflow can continue or recover appropriately.

Final Answer: C

C = Report unresolved failures with attempted section, failure type, partial fields, and recovery suggestions — this preserves failure context and allows reliable downstream handling without aborting the entire document.

Question: 15 CertyIQ

A multi-agent research system occasionally generates reports while the synthesis agent is still incomplete, resulting in missing or unverifiable citations.
What change prevents invalid reports?

A. Generate reports immediately and remove invalid citations afterward.
B. Tell the report agent to refuse incomplete synthesis.
C. Check whether synthesis messages contain words like “complete.”
D. Block report-generation tasks until synthesis returns a structured citation manifest and success status.

Answer: D

Explanation:

A. Generate reports immediately and remove invalid citations afterward.

Incorrect

This allows incomplete reports to be generated. Removing citations afterward does not guarantee that the remaining evidence is complete or verifiable.

B. Tell the report agent to refuse incomplete synthesis.

Incorrect

This relies on the report agent correctly recognizing incomplete input. It is better to enforce the dependency before the report-generation task starts.

C. Check whether synthesis messages contain words like "complete."

Incorrect

Keyword checks are unreliable. A message can say "complete" while still missing required citations or evidence.

D. Block report-generation tasks until synthesis returns a structured citation manifest and success status.

Correct

This creates an explicit **workflow gate**. The report cannot start until the synthesis agent provides a structured citation manifest and confirms successful completion. This ensures the report is based on complete and verifiable evidence.

Final Answer: D

D = Block report generation until the synthesis agent returns a structured citation manifest and success status, preventing incomplete or unverifiable reports from being produced.

Question: 16 CertyIQ

A migration assistant sometimes starts dependency analysis before collecting required package manager and framework versions.
What change most reliably enforces the correct first step?

A. Detect whether Claude mentions metadata completion in natural language.
B. Force the metadata extraction tool with tool_choice for the first request.
C. Set tool_choice to any for the first request.
D. Strengthen the system prompt requiring metadata extraction first.

Answer: B

Explanation:

A. Detect whether Claude mentions metadata completion in natural language.

Incorrect

Natural-language detection is unreliable. Claude may mention that metadata is complete without actually calling the required extraction tool or obtaining all necessary information.

B. Force the metadata extraction tool with tool_choice for the first request.

Correct

Using tool_choice explicitly forces the required metadata extraction tool to run first. This provides a deterministic workflow and prevents dependency analysis from starting before the required package manager and framework versions are collected.

C. Set tool_choice to any for the first request.

Incorrect

any allows the model to choose **any available tool**. It does not guarantee that the metadata extraction tool will be selected first.

D. Strengthen the system prompt requiring metadata extraction first.

Incorrect

A stronger prompt can guide the model, but it does not reliably enforce tool execution order. Explicit tool selection provides a stronger control mechanism.

Final Answer: B

B = Force the metadata extraction tool with tool_choice for the first request, ensuring the required metadata is collected before dependency analysis begins.

Question: 17 CertyIQ

A team wants Claude-generated pull request reviews to integrate with existing development systems. What output approach should be selected?

A. Free-form text responses
B. Structured JSON with validation rules
C. Long markdown explanations only
D. Unformatted comments

Answer: B

Explanation:

A. Free-form text responses

Incorrect

Free-form text can vary in structure and wording, making it difficult for automated development systems to reliably parse and process the review results.

B. Structured JSON with validation rules

Correct

Structured JSON provides a **consistent, machine-readable format**. Validation rules or a defined schema ensure required fields, data types, and structure remain consistent, making the output easier to integrate with existing PR and development tools.

C. Long markdown explanations only

Incorrect

Markdown is useful for human-readable reviews but is less reliable for automated processing because formatting and content can vary.

D. Unformatted comments

Incorrect

Unformatted comments lack a predictable structure and make automated integration and downstream processing difficult.

Final Answer: B

B = Structured JSON with validation rules ensures consistent, machine-readable PR review output that can be reliably integrated with existing development systems.

Question: 18 CertyIQ

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Free-form text can vary in structure and wording, making it difficult for automated development systems to reliably parse and process the review results.

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Structured JSON provides a **consistent, machine-readable format**. Validation rules or a defined schema ensure required fields, data types, and structure remain consistent, making the output easier to integrate with existing PR and development tools.

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Incorrect

Markdown is useful for human-readable reviews but is less reliable for automated processing because formatting and content can vary.

D. Unformatted comments

Incorrect

Unformatted comments lack a predictable structure and make automated integration and downstream processing difficult.

Final Answer: B

B = Structured JSON with validation rules ensures consistent, machine-readable PR review output that can be reliably integrated with existing development systems.

Question: 19 CertyIQ

A developer wants Claude to return API responses that can be directly consumed by another application.
Which technique should be used?

A. Request formatted paragraphs
B. Use structured output with defined fieldscorrect
C. Ask Claude for longer explanations
D. Allow unrestricted responses

Answer: B

Explanation:

A. Request formatted paragraphs

Incorrect

Paragraphs are designed for human readability and do not provide a predictable structure that another application can reliably parse.

B. Use structured output with defined fields

Correct

Structured output with defined fields provides a **consistent, machine-readable response format**. This allows another application to reliably parse and consume the API response.

C. Ask Claude for longer explanations

Incorrect

Longer explanations add more natural-language content but do not guarantee a predictable structure for automated processing.

D. Allow unrestricted responses

Incorrect

Unrestricted responses can vary in format and content, making reliable application-to-application integration difficult.

Final Answer: B

B = Use structured output with defined fields to produce consistent, machine-readable API responses that another application can consume directly.

Question: 20 CertyIQ

Multiple backend problems are returned to an agent as the same message: “Operation failed.” Problems include invalid order IDs, temporary service timeouts, expired tokens, and policy restrictions.
What change most improves recovery behavior?

A. Retry every failed tool call twice before escalation.
B. Keep generic failures and let Claude infer retryability.

C. Map failures to customer sentiment levels.

D. Return structured MCP errors with categories, retryability, and user-safe details.

Answer: D

Explanation:

A. Retry every failed tool call twice before escalation.

Incorrect

Not every failure is retryable. Retrying invalid order IDs or policy restrictions wastes resources and can make failures worse.

B. Keep generic failures and let Claude infer retryability.

Incorrect

A generic error message does not provide enough reliable information for the agent to determine whether a failure should be retried, corrected, or escalated.

C. Map failures to customer sentiment levels.

Incorrect

Customer sentiment does not determine the technical recovery strategy. The agent needs actionable information about the actual failure.

D. Return structured MCP errors with categories, retryability, and user-safe details.

Correct

Structured MCP errors give the agent **explicit recovery information**. For example, a timeout can be marked retryable, while an invalid order ID or policy restriction can be marked non-retryable. User-safe details also allow the agent to communicate an appropriate explanation without exposing sensitive backend information.

Final Answer: D

D = Return structured MCP errors containing failure categories, retryability, and user-safe details so the agent can choose the appropriate recovery action.

Question: 21 CertyIQ

Scenario: Multi-Agent Research System
You are building a multi-agent research system using the Claude Agent SDK. A coordinator agent delegates to specialized subagents: one searches the web, one analyzes documents, one synthesizes findings, and one generates reports.
Your coordinator's document-analysis subagent depends on an internal library-catalog MCP server. It works in your local Claude Code session, but two teammates who cloned the repository cannot access the catalog tools, so their runs fall back to weaker web searches. Separately, one researcher is testing an experimental notes server that should not affect the team.
What should you change?

A. Document per-user claude mcp add commands in README, and ask each researcher to recreate both servers locally.

B. Place both servers in each user's ~/.claude.json, and rely on onboarding scripts to synchronize changes across machines.

C. Combine the catalog and notes integrations into one MCP server, and expose only role-specific tools to subagents.

D. Move the required catalog server into project-scoped .mcp.json, and keep the experimental notes server in user-scoped ~/.claude.json.

Answer: D

Explanation:

A. Document per-user claude mcp add commands in README, and ask each researcher to recreate both servers locally.

Incorrect

This creates manual setup and configuration drift. It also does not provide a shared, version-controlled configuration for the required team integration.

B. Place both servers in each user's ~/.claude.json, and rely on onboarding scripts to synchronize changes across machines.

Incorrect

User-scoped configuration is intended for personal integrations. Putting the required catalog server there makes the team dependency dependent on each user's local configuration.

C. Combine the catalog and notes integrations into one MCP server, and expose only role-specific tools to subagents.

Incorrect

Combining unrelated integrations is unnecessary and reduces configuration isolation. The experimental notes server should remain separate from the team's required catalog dependency.

D. Move the required catalog server into project-scoped .mcp.json, and keep the experimental notes server in user-scoped ~/.claude.json.

Correct

Project-scoped .mcp.json is appropriate for **team-shared MCP configuration** that should travel with the repository. This ensures teammates receive the required catalog server configuration when they clone the project.

The experimental notes server is personal and should remain in ~/.claude.json, keeping it isolated from the shared team configuration.

Final Answer: D

D = Put the required catalog MCP server in project-scoped .mcp.json for consistent team access, while keeping the experimental notes server user-scoped in ~/.claude.json.

Question: 22 CertyIQ

Scenario: Developer Productivity with Claude
A Claude-powered helper updates legacy configuration files to a new format. Initial generations look reasonable, but repeated runs keep breaking different corner cases:
l Missing optional sections

l Duplicate keys
l Large files with thousands of entries
Developers have been sending prompts like "fix the migration bugs" and receiving inconsistent improvements.
What refinement approach should they use next?

A. Ask Claude to reread the entire legacy codebase and rewrite the migration helper from scratch after each failed run.

B. Remove the most difficult corner-case checks temporarily so Claude can stabilize the happy-path migration behavior first.

C. Tell Claude the output is still wrong, request a more careful implementation, and continue until reviewers approve it.

D. Share the failing test names, inputs, expected versus actual outputs, and stack traces, then iterate on minimal fixes.

Answer: D

Explanation:

A. Ask Claude to reread the entire legacy codebase and rewrite the migration helper from scratch after each failed run.

Incorrect

Rewriting the entire helper after every failure is inefficient and can introduce new regressions. It does not isolate the specific causes of the failures.

B. Remove the most difficult corner-case checks temporarily so Claude can stabilize the happy-path migration behavior first.

Incorrect

Removing important checks only hides the underlying problems. The migration tool still needs to correctly handle optional sections, duplicate keys, and large files.

C. Tell Claude the output is still wrong, request a more careful implementation, and continue until reviewers approve it.

Incorrect

This provides vague feedback without giving Claude enough information to identify the actual failure. Repeatedly asking for a “more careful” implementation can lead to inconsistent changes.

D. Share the failing test names, inputs, expected versus actual outputs, and stack traces, then iterate on minimal fixes.

Correct

This provides **specific, actionable debugging evidence**. Claude can compare the expected and actual behavior, identify the exact failure, and make a targeted change rather than rewriting unrelated code.

Iterating on **minimal fixes** also reduces the risk of introducing new regressions while progressively addressing each corner case.

Final Answer: D

D = Provide concrete failure evidence — test names, inputs, expected/actual results, and stack traces — and iterate with minimal targeted fixes.

Question: 23 CertyIQ

Supplier contract extraction returns null values for effective dates and renewal terms even though information exists in cover letters, footnotes, and table captions.
What change most improves accuracy?

A. Retry until all metadata fields contain values.
B. Make fields required and non-null.
C. Add targeted few-shot examples showing extraction from varied document sections.
D. Tell Claude null values are unacceptable for supplier contracts.

Answer: C

Explanation:

A. Retry until all metadata fields contain values.

Incorrect

Retries do not solve the underlying extraction problem. If the model consistently overlooks information in certain document sections, repeated attempts may produce the same result.

B. Make fields required and non-null.

Incorrect

Making fields mandatory does not teach the model **where or how to find the information**. It could also encourage the model to produce unsupported values instead of correctly returning null when information is genuinely absent.

C. Add targeted few-shot examples showing extraction from varied document sections.

Correct

Targeted few-shot examples demonstrate that relevant information can appear in **cover letters, footnotes, table captions, and other non-standard sections**. This gives Claude concrete examples of the expected extraction behavior and improves consistency across varied document layouts.

D. Tell Claude null values are unacceptable for supplier contracts.

Incorrect

This only discourages null values without providing a reliable extraction strategy. It may cause the model to guess or hallucinate values when information cannot actually be found.

Final Answer: C

C = Add targeted few-shot examples demonstrating extraction from varied document sections, helping Claude recognize and extract contract information even when it appears outside the main body.

Question: 24 CertyIQ

A legacy subscription service has cancellation behavior distributed across command-line scripts, web handlers, and scheduled jobs. Previous direct edits often targeted the wrong abstraction.
Which workflow should be recommended first?

A. Read every repository file before implementing changes.
B. Use plan mode to map relevant flows and compare implementation points before modifications.
C. Directly edit and rely on test failures to guide corrections.
D. Start separate direct sessions for each module and combine changes later.

Answer: B

Explanation:

A. Read every repository file before implementing changes.

Incorrect

Reading the entire repository is inefficient and can overwhelm the analysis with irrelevant information. The goal is to identify the relevant flows and implementation points first.

B. Use plan mode to map relevant flows and compare implementation points before modifications.

Correct

Plan mode allows Claude to **investigate the codebase, trace the cancellation behavior, identify related components, and compare possible implementation points** before making changes. This is especially useful when behavior is distributed across multiple abstractions.

C. Directly edit and rely on test failures to guide corrections.

Incorrect

This approach can lead to repeated incorrect edits and unnecessary regressions. Understanding the architecture first reduces the chance of modifying the wrong layer.

D. Start separate direct sessions for each module and combine changes later.

Incorrect

Separate sessions can lack shared context and may produce inconsistent approaches across modules. A coordinated plan provides a clearer understanding of the complete workflow.

Final Answer: B

B = Use plan mode to map the relevant flows and compare implementation points before making modifications, reducing the risk of editing the wrong abstraction.

Question: 25 CertyIQ

A team builds an application where Claude must classify customer requests into predefined categories. Which approach provides the most consistent results?

A. Provide examples of each category and expected classifications
B. Ask Claude to create new categories when needed
C. Use a random classification approach.
D. Remove category definitions from the prompt

Answer: A

Explanation:

A. Provide examples of each category and expected classifications

Correct

Providing clear category definitions along with **examples of inputs and their expected classifications** helps Claude understand the intended boundaries between categories. This few-shot approach improves consistency and reduces ambiguous classifications.

B. Ask Claude to create new categories when needed

Incorrect

This can cause inconsistent results because Claude may introduce categories outside the predefined classification scheme.

C. Use a random classification approach

Incorrect

Random classification cannot provide reliable or consistent results and does not use the defined category criteria.

D. Remove category definitions from the prompt

Incorrect

Without category definitions, Claude lacks the necessary guidance to determine which predefined category best matches each request.

Final Answer: A

A = Provide examples of each category and expected classifications to establish clear patterns and produce more consistent classification results.

Question: 26 CertyIQ

Scenario: Customer Support Resolution Agent
You are building a customer support resolution agent using the Claude Agent SDK. The agent handles high-ambiguity requests like returns, billing disputes, and account issues. It has access to backend systems through MCP tools (get_customer, lookup_order, process_refund, escalate_to_human). Your target is 80%+ first-contact resolution while knowing when to escalate.
During billing disputes, the agent creates condensed case notes before responding or escalating. Human reviewers report that notes sometimes say things like "refund approved" or "customer was promised free return shipping," but they cannot tell whether each claim came from the customer, an order record, a policy document, or the agent's inference. In several escalations, customer allegations were treated as verified backend facts.
What change would most effectively reduce this reliability risk?

A. Compress each case into a shorter narrative summary and instruct reviewers to ask follow-up questions when provenance seems unclear.

B. Store complete raw transcripts and tool outputs only, relying on human reviewers to reconstruct sources when disputes arise.

C. Require summaries and handoffs to carry structured claim-source mappings with source type, excerpt or tool field, timestamp, and verification status.

D. Ask the agent to include only high-confidence claims in final responses, omitting uncertain details from summaries and escalations.

Answer: C

Explanation:

A. Compress each case into a shorter narrative summary and instruct reviewers to ask follow-up questions when provenance seems unclear.

Incorrect

Shorter summaries can make provenance even harder to determine. Relying on reviewers to reconstruct unclear information introduces unnecessary manual effort and increases the chance of misinterpreting claims.

B. Store complete raw transcripts and tool outputs only, relying on human reviewers to reconstruct sources when disputes arise.

Incorrect

Raw data preserves the evidence but does not make the relationship between each claim and its source explicit. Reviewers would still need to manually reconstruct the provenance.

C. Require summaries and handoffs to carry structured claim-source mappings with source type, excerpt or tool field, timestamp, and verification status.

Correct

Structured **claim provenance** makes it clear where every important statement originated and whether it has been verified. For example, a claim can be explicitly associated with a customer statement, backend field, policy document, or agent inference.

Including the **source type, evidence excerpt/tool field, timestamp, and verification status** gives human reviewers enough context to distinguish customer allegations from verified system facts.

D. Ask the agent to include only high-confidence claims in final responses, omitting uncertain details from summaries and escalations.

Incorrect

Removing uncertain claims can hide important information needed for escalation. The issue is not simply confidence; it is the inability to distinguish **verified facts from allegations and inferences**.

Final Answer: C

C = Require structured claim-source mappings with evidence, timestamps, and verification status so every claim can be traced to its origin and distinguished from unverified allegations or agent inference.

Question: 27 CertyIQ

A research system creates coherent reports, but coverage gaps remain. For example, AI regulation reports have strong EU coverage but weak Asia-Pacific coverage.
What change improves completeness?

A. Repeat broad assignments twice.
B. Hide unsupported sections.
C. Increase coordinator iteration count to a fixed number.
D. Add a coordinator review loop that checks coverage, delegates follow-ups, and reruns synthesis.

Answer: D

Explanation:

A. Repeat broad assignments twice.

Incorrect

Repeating the same broad assignment does not guarantee that the missing Asia-Pacific coverage will be identified or addressed. It may simply produce similar results.

B. Hide unsupported sections.

Incorrect

Hiding unsupported sections reduces transparency and does not solve the underlying coverage gap.

C. Increase coordinator iteration count to a fixed number.

Incorrect

A fixed number of iterations does not ensure that important gaps have been resolved. The system needs to evaluate **what is missing**, not simply perform more iterations.

D. Add a coordinator review loop that checks coverage, delegates follow-ups, and reruns synthesis.

Correct

A coordinator review loop can explicitly **evaluate coverage against the research requirements**, identify gaps such as weak Asia-Pacific coverage, delegate targeted follow-up research, and then rerun synthesis with the new findings.

This creates an adaptive feedback loop rather than relying on repeated broad searches.

Final Answer: D

D = Add a coordinator review loop that checks coverage, delegates targeted follow-ups, and reruns synthesis to close identified research gaps.

Question: 28 CertyIQ

A developer wants Claude to review code and return only actionable issues. What prompt design is most effective?

A. Request every possible improvement
B. Define what qualifies as an issue and provide examples.
C. Ask for unlimited suggestions
D. Avoid describing expected output

Answer: B

Explanation:

A. Request every possible improvement

Incorrect

This can produce excessive suggestions, including minor style preferences or changes that are not actually

actionable. It makes the review less focused.

B. Define what qualifies as an issue and provide examples

Correct

Clearly defining what counts as an **actionable issue** gives Claude precise evaluation criteria. Examples further demonstrate the expected severity, scope, and output behavior, leading to more consistent reviews.

C. Ask for unlimited suggestions

Incorrect

Unlimited suggestions can create noisy reviews and make it harder for developers to identify the issues that genuinely require action.

D. Avoid describing expected output

Incorrect

Without clear output requirements, Claude may produce inconsistent review formats or include unnecessary commentary.

Final Answer: B

B = Define what qualifies as an actionable issue and provide examples, giving Claude clear criteria for focused and consistent code reviews.

Question: 29 CertyIQ

Scenario: Customer Support Resolution Agent
Production logs show:
l get_customer returns ISO 8601 timestamps and numeric loyalty tiers.
l lookup_order returns Unix timestamps and numeric status codes.
The agent sometimes miscomputes return windows or mislabels order status.
You cannot modify upstream MCP servers.
What change best addresses the reliability issue?

A. Escalate all cases with mixed timestamp formats.

B. Add a PostToolUse hook that converts tool results into canonical dates, status labels, and tier fields before Claude sees it.

C. Add system instructions telling Claude to remember each tool's format

D. Modify the final customer response parser.

Answer: B

Explanation:

A. Escalate all cases with mixed timestamp formats.

Incorrect

Escalating every case avoids the problem rather than fixing it. It would unnecessarily increase human workload and reduce the agent's ability to resolve cases automatically.

B. Add a PostToolUse hook that converts tool results into canonical dates, status labels, and tier fields before Claude sees it.

Correct

A **PostToolUse hook** can normalize inconsistent tool outputs before they are processed by Claude. Converting timestamps, status codes, and tier values into consistent canonical representations reduces interpretation errors and makes downstream reasoning more reliable.

C. Add system instructions telling Claude to remember each tool's format.

Incorrect

Instructions can help, but they rely on Claude consistently remembering and correctly interpreting different formats. Deterministic normalization at the tool-result boundary is more reliable.

D. Modify the final customer response parser.

Incorrect

The errors occur during the agent's reasoning and calculation of return windows/statuses. Changing only the final response parser happens too late to correct the underlying interpretation.

Final Answer: B

B = Add a PostToolUse hook to normalize tool results into canonical dates, status labels, and tier fields before Claude processes them, improving reliability without modifying the upstream MCP servers.

Question: 30 CertyIQ

A multi-agent research system assigns high confidence to most report claims, but customer audits find citation mismatches concentrated in regulatory topics and newly added source types.
What review strategy should be implemented before further reducing reviewer coverage?

A. Review only claims with low confidence scores.

B. Use one aggregate accuracy metric and reduce review after reaching the target.

C. Auto-publish high-confidence reports and investigate only customer defects.

D. Randomly sample high-confidence claims across topic, source type, and report section, then track errors by category.

Answer: D

Explanation:

A. Review only claims with low confidence scores.

Incorrect

Confidence scores alone are not sufficient. The audits already show that **high-confidence claims can contain citation errors**, particularly in specific topics and source types.

B. Use one aggregate accuracy metric and reduce review after reaching the target.

Incorrect

An aggregate metric can hide errors concentrated in specific categories. Overall accuracy may look good while regulatory claims continue to have poor citation quality.

C. Auto-publish high-confidence reports and investigate only customer defects.

Incorrect

This is reactive rather than preventive. High-confidence reports may still contain systematic citation mismatches, so relying only on customer-reported defects leaves important errors undetected.

D. Randomly sample high-confidence claims across topic, source type, and report section, then track errors by category.

Correct

Stratified sampling helps detect **hidden errors within high-confidence outputs**. Sampling across topic, source type, and report section is particularly important because the existing defects are concentrated in regulatory topics and newly added sources.

Tracking errors by category also reveals where additional reviewer coverage or targeted improvements are needed before reducing human review.

Final Answer: D

D = Randomly sample high-confidence claims across important dimensions and track errors by category to detect systematic weaknesses before reducing reviewer coverage.

Question: 31

CertyIQ

Scenario: Structured Data Extraction
A labeled validation run reports:
l 97.2% aggregate extraction accuracy
l coverage across NDAs, invoices, insurance forms, and purchase orders
Leadership wants to bypass human review for any record where Claude returns high confidence, but pilot complaints suggest errors cluster in specific document families and fields.
What should you do before reducing reviewer coverage?

A. Analyze accuracy by document type and extracted field, then automate only segments meeting validated thresholds while routing others to review.

B. Randomly review a fixed percentage of every extraction category, regardless of document type, field, or prior validation performance.

C. Raise the global confidence threshold until the aggregate validation accuracy reaches 99%, then disable review for remaining outputs.

D. Use the 97.2% aggregate accuracy to auto-approve all high-confidence outputs, monitoring user complaints for emerging regressions.

Answer: A

Explanation:

A. Analyze accuracy by document type and extracted field, then automate only segments meeting validated thresholds while routing others to review.

Correct

Aggregate accuracy can hide **localized weaknesses**. A 97.2% overall score may look strong while particular document types or fields have significantly higher error rates.

Breaking accuracy down by **document type and extracted field** identifies these weak segments. Automation can then be safely applied only where validation shows acceptable performance, while higher-risk segments continue receiving human review.

B. Randomly review a fixed percentage of every extraction category, regardless of document type, field, or prior validation performance.

Incorrect

Uniform sampling ignores known differences in performance. Higher-risk document families and fields may require more focused validation rather than the same review percentage everywhere.

C. Raise the global confidence threshold until the aggregate validation accuracy reaches 99%, then disable review for remaining outputs.

Incorrect

Increasing a global confidence threshold does not address **systematic errors concentrated in specific fields or document types**. High confidence can still be poorly calibrated for certain segments.

D. Use the 97.2% aggregate accuracy to auto-approve all high-confidence outputs, monitoring user complaints for emerging regressions.

Incorrect

This relies on aggregate performance and reactive customer feedback. It does not adequately validate whether high-confidence outputs are reliable within the specific segments where errors are already concentrated.

Final Answer: A

A = Analyze accuracy by document type and field, then automate only validated high-performing segments while keeping higher-risk segments under human review.